Exp.no:1 Basic Windows / Linux Networking Commands

Windows:-

1. IP Configuration
   - ipconfig : Displays basic network details

3.ipconfig

```
C:\Windows\system32>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::bc96:2f44:bfbf:5a00%3
   IPv4 Address. . . . . . . . . . . : 172.16.9.19
   Subnet Mask . . . . . . . . . . . : 255.255.252.0
   Default Gateway . . . . . . . . . : 172.16.8.1
```

- ipconfig /all : shows detailed network adapter information including MAC address, DNS servers & DHCP status

```
C:\Windows\system32>ping google.in

Pinging google.in [142.250.207.68] with 32 bytes of data:
Reply from 142.250.207.68: bytes=32 time=3ms TTL=119
Reply from 142.250.207.68: bytes=32 time=3ms TTL=119
Reply from 142.250.207.68: bytes=32 time=3ms TTL=119
Reply from 142.250.207.68: bytes=32 time=3ms TTL=119

Ping statistics for 142.250.207.68:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 3ms, Average = 3ms
```

• ipconfig /release : Releases the current IP address assigned by DHCP server.

```
C:\Windows\system32>tracert google.in

Tracing route to google.in [142.250.207.68]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  172.16.8.1
  2     1 ms     1 ms     2 ms  137.97.163.129
  3     *        *        *     Request timed out.
  4     4 ms     4 ms     5 ms  172.16.12.64
  5     3 ms     3 ms     3 ms  172.16.12.64
  6     5 ms     4 ms     4 ms  72.14.217.252
  7     5 ms     3 ms     4 ms  142.251.227.213
  8     3 ms     4 ms     3 ms  142.251.55.205
  9     3 ms     3 ms     3 ms  hkg12s32-in-f4.1e100.net [142.250.207.68]

Trace complete.
```

2. Connectivity & Path Troubleshooting.

• ping <destination> : Tests connectivity between your computer & another device by sending ICMP echo requests.

6.. Hostname

```
C:\Windows\system32>hostname
DESKTOP-0H11URO
```

requests until stopped manually.

- tracert <destination>: Displays the path packets take to reach a destination & identifies delays.

- pathping <destination>: Combined ping & tracert to analyse packet loss & network latency.

3. Ports & connections:

- netstat -ano: Displays all active network connections listening ports & associated process IDs

- netstat -p tcp: To get information about status of TCP/IP network routes

4. DNS cache Management

- ipconfig /flushdns: Clear local DNS cache